NAMES: RAMBOFHENI VHUHWAHO

EXERCISE 2: SQL fundamentals

Database: employees_db

Questions

① COUNT() function

Write a SQL query to find the total number of employees in the company.

```
SELECT COUNT (DISTINCT id)
FROM employees_db
```

id employees_db

Id
1
2
3
4
5
6
7
8
9
10

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② SUM() Function

Write a SQL query to find the total Salary Paid to all employees in the IT department.

```
SELECT sum (~~Tutot~~_salary) AS Tutal_Salary
FROM employees_db
~~GROUP BY IT department~~;
WHERE department = 'IT';
```

employees_db

department	Tutal_Salary
IT	220 000

③ AVG() Function

Write a SQL query to Calculate the average Salary of employees in the HR department

```
SELECT AVG (salary) AS Average_Salary
FROM employees_db
WHERE department = 'HR';
```

employees_db

department	Average_Salary
HR	

④ MIN and MAX Functions

Write a SQL query to find the highest and lowest Salary in the company.

```
SELECT MAX (salary) AS highest_Salary
~~FROM Salary~~;
FROM employees_db
```

employee_db

highest_Salary
62000

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④ MIN and MAX FUNCTION ....Continuation of Question 4

```
SELECT MIN(salary) AS lowest_salary
FROM employees_db
```

employee_db

lowest_salary
50 000

⑤ GROUP BY Statement

Write a SQL query to group employees by department and display the total salary paid in each department.

```
SELECT SUM(salary) AS Total_salary,
        ~~first_name~~,
        ~~last_name~~,
        department
FROM employees_db
GROUP BY department
```

employees_db

department	Total_salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

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⑥ GROUP BY and COUNT()

write a SQL query to count how many employees work in each city.

```
SELECT COUNT(DISTINCT ID),
        City
FROM employees_db
GROUP BY city
```

employees_db

Id	City
4	IT
2	HR
2	Finance
2	Marketing

⑦ GROUP BY and ORDER BY

write a SQL query to group employees by department, calculate the average Salary in each department and order the results in descending order of average Salary.

⑧ HAVING Clause

Write a SQL query to find departments where the total salary paid exceeds 100 000 (use GROUP BY and HAVING)

```
SELECT SUM (salary) AS total_salary,
       department
FROM employees_db
GROUP BY department;
HAVING 'total_salary' > 100 000
```

employees_db

department	Total_salary
IT	220 000
finance	119 000
Marketing	105 000

⑨ Combining GROUP BY, HAVING, and ORDER BY

Write a SQL Query to list cities where more than one employee works, ordered by the number of employees in descending order,

```
SELECT COUNT(DISTINCT id),
       city
FROM employees_db
GROUP BY city
~~HAVING employees DESC~~
~~ORDER BY~~
~~Having~~ HAVING number called of employees ~~DESC~~
ORDER BY ~~id~~ DESC
HAVING COUNT (10) number of employee
ORDER BY id DESC
```

employees_db

id	city
3	Chicago
2	New York
2	Los Angeles
2	San francisco

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